



TM3 Review Guide for Test 3 on 03.12

Completion requirements

Test format. The test format will be similar to that of Test 2, with a mix of objective questions (e.g., multiple choice) and problem solving.

As always, please review the slides and notes for all content covered since the last quiz. Keep in mind that we sometime use the board, so not everything is included on a slide.

Machine Learning Overview

- There are many words that you should know well enough to identify its in context or use in a sentence.
 - **Regression** vs **classification** (Note: linear regression is just one form of regression!)
 - **Supervised** and **unsupervised learning**
 - **Features** and **class label**
 - **Binary** and **multi-class classification** problems
 - **Feature space**
 - **Decision boundary** (or **decision surface**)
 - **Fitting a model** to the data

Logistic Regression

- More words to know:
 - **Logistic regression**
 - **Linear model** vs. non-linear model
 - **Probability, odds, log odds (logit)**
 - **Binary** vs **multinomial logistic regression**
 - Why is the term *logistic regression* a misnomer? (In class we discussed that the word "regression" in this case does not suggest regression learning.)
- Be prepared to convert between probability, odds, and log odds
- You should know the **logit** and **logistic functions** by heart and have an intuitive understanding of how they relate, what their graphs look like, and what they represent.
- What scenario is best suited to logistic regression? Compare and contrast this to the scenario in which linear regression is best suited.
- Given a logistic regression model, be prepared to:
 - Compute the probability of a binary event given the features of the instance (compute z , then p)
 - Classify the instance

- What are some of the advantages and disadvantages of logistic regression?
- There will not be any questions asking you how to use `sklearn.linear_model.LogisticRegression` or similar packages. However, you should be prepared to explain the **coefficients** (weights **w**) and intercept w_0 that the method learns from the data and why, for multinomial problems, there is one row (vector) per class label.
- You may be shown a Pandas dataframe or a visualization of feature space and asked to point out features, class label, decision boundary, etc., or asked to explain other aspects of it from a practical standpoint.

How to Evaluate a Model

Model Building and Evaluation (Train-Test Process)

In these two sessions, we discussed the general procedure for building and evaluating machine learning models. You should be familiar with each of these steps and understand how they contribute to evaluating a model's performance:

1. **Randomly split data** into training and test sets (train-test split).
2. **Initialize the chosen classifier** (e.g., logistic regression, kNN).
3. **Fit the model to the training data** (learn model parameters).
4. **Use the trained model** to make predictions on the test set.
5. **Evaluate model performance** using metrics (accuracy, precision, recall, F_1 score, confusion matrix).

Note: Evaluation specifically refers to measuring how well your trained model performs on unseen (test) data.

- There are many terms to know:
 - **train-test split**
 - **i.i.d.**
 - **(simple) random sampling** and **stratified sampling**
 - **overfitting** and **underfitting**
 - **bias error and variance**
 - **confusion matrix**
 - **true** and **false positive**, **true** and **false negative**
 - **accuracy**
 - **precision** and **recall**
 - **F_1 score** (the **harmonic mean** of precision and recall)
 - Note: The harmonic mean is just the reciprocal of the mean of the reciprocals. So if precision is P and recall is R, then the harmonic mean

(f1-score) of P and R is just $2(1/P + 1/R)$. Just so you know where that comes from.

- **support** (the number of samples of the true response that lie in that class, so in the example code for the IRIS test set, the test set had 10 of each class, so the support in each case was 10) [see F.A.Q. below for more details]
- Note: We "overload" certain terms in many fields. Be careful not to confuse the *bias* (intercept) w_0 of a linear model with the *bias error* that results from undertraining. These concepts are *not* directly related. Context will usually tell you which meaning is intended.
- Further note: Accuracy is often reported as a percent. But I very rarely see f1 scores reported in this way, and if you think about it, they aren't really a percent. So just report f1 as a decimal value. Precision and recall are a bit of a toss-up. I think you could report those either as a percent or a decimal, just so long as you don't get that mixed up (0.25 is 25% not 0.25%). And support is just a count, so it is usually reported as an integer.
- **You should expect a problem with text describing in words how a model performs on a test set.** From that description, you should be prepared to:
 - construct a confusion matrix with properly labeled rows and columns;
 - compute metrics such as accuracy, precision, recall, f1 score, and support;
 - discuss in general terms the suitability of the model given the problem.
 - **Conversely, you may be given a confusion matrix** (it could be in different forms—for a multi-class problem, output from sklearn, etc.) **and asked to compute the stats described above and to evaluate the model in words.**
- Suppose that, as a practicing data scientist, you have fit a logistic regression model to a data set that is important to your company. You have carefully evaluated your model, and all of the results look promising to you. However, your boss is not yet convinced. It will be an expensive rollout, costing millions of dollars. "Why do you think your model will work for us in the field?" your boss asks. Explain the steps that you have taken, and why the company can have confidence in your model. Make sure your answer is clear and convincing. (Your boss doesn't know much about statistics or machine learning, but she is a "hell of an engineer" and she is sure to see straight through any B.S. you try to offer up in defense of your model.)

k-Nearest Neighbors (kNN)

- You should be familiar with the following terms and able to use them in a sentence:
 - **parametric method**
 - **nonparametric method**
 - **distance metric**
 - **parameter** (in the context of statistics or supervised machine learning)
 - **hyperparameter**

- **neighborhood**
- **k-nearest neighbors (kNN) method**
- **1-nearest neighbor (1NN) method** (special case of kNN: $k=1$)
- You should be able to **give examples of parametric** and **nonparametric** machine learning methods.
- You should be able to **give examples of parameters** for all of the methods that we discuss in class (including those in the future!)
- You should be able to **give examples of hyperparameters** for all of the methods that we discuss in class (including those in the future!)
- Make sure you are familiar with the common distance metrics and know how to compute them with a calculator if given a simple problem with 3-4 features:
 - **Euclidean** (L_2 distance)
 - **Manhattan** (L_1) distance
 - **Chebyshev** (L_∞) (max) distance
 - **Hamming** distance
 - You should know that there are MANY different distance metrics for special-purpose applications. (Two of my favorite are the **Haversine** distance for estimating the distance between two geolocations on a globe and the **Kendall tau rank distance** for computing the distance between two rankings. The latter is useful in preference learning and voting theory. But I probably wouldn't ask you about either of these on a test!)
- You should be familiar with the **k-nearest neighbors methods** for classification and regression.
- You should know that these methods depend on suitable application of **scaling** during preprocessing.
- You should know that we sometimes use methods like **grid search** and a **validation set** to find optimal values of hyperparameters—for example, the best value of k and the best distance metric for kNN. (We may or may not have covered this yet in class. If not, it won't be on the quiz. But it's an important concept, so you should go ahead and look it up!)

Note: There will be no questions on Decision Trees since we only covered this on Monday. It will be covered on the next test on Wednesday after Spring Break.

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